

29. INTEGRATION OF THE EXTERNAL COELOM INTO AN INTERNAL ONE

DURATION : 08:40

STUDY SHEET

COURSE SUMMARY

This video explores the complex process of external coelome integration into an internal coelome, essential for embryonic development. It explains how the peritoneum forms in parallel with the digestive tract and how mechanical and genetic environmental influences shape our epigenetics. The concepts of polarity, notochord, and motility are presented, illustrating how the amniotic cavity and aortas interact to create the embryonic architecture. By analysing this process, the video highlights the importance of the peritoneum not only as a protective structure but also as a key player in metabolic and genetic integration, ensuring our adaptation to the environment from the embryonic stage.

INTEGRATION OF THE EXTERNAL COELOM INTO AN INTERNAL COELOM

The **peritoneum** develops simultaneously with the developmental phase of the integration of the **external coelom** into an **internal coelom**. This primitive establishment of the peritoneum begins with the polarity and information on the **neural tube**, whose growth is organised in space and time by the fluidic reorganisation of the embryo, particularly the early development of the **primitive aortae**. This process is a constant quest to return to the midline, with the aorta serving as this return axis.

The peritoneum responds to this development and is its expression. We observe the **amniotic cavity**, the **vitelline cavity**, and the **embryonic pedicle**, surrounded by the external coelom. In the next developmental phase, the external coelom integrates into an internal coelom synchronously with the establishment of the primitive digestive tract, as well as the visceral and parietal peritoneum. Between these two peritonea, a fluid originating from the external coelom integrates to become the **intraperitoneal fluid**. It is important to note that there are no intraperitoneal organs, only peritonealised organs, with a few exceptions in women, where an opening between the peritoneum and the external world occurs, notably with the **fallopian tubes**.

The uterus is a concentration of parietal peritoneum, where the fallopian tubes condense and form an opening to the peritoneal cavity. The growth of the whole is significant, and everything organises and delimits itself. To better understand, a cross-section can be made.

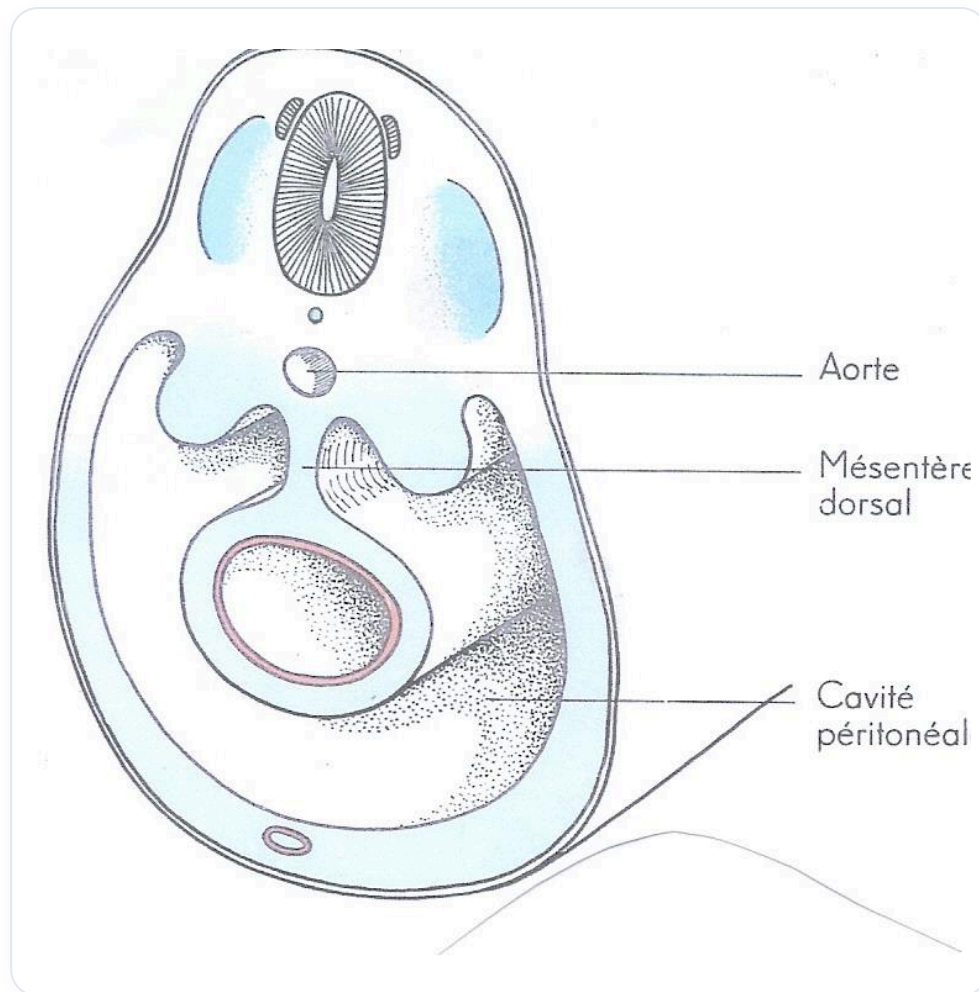


FIG. — Embryo cross-section

Turning this cross-section towards us, we observe the amniotic cavity and the vitelline cavity. The **notochordal process** is a subtle driver for development, while the formation of the neural tube guides growth. This tube, through its expansion, is guided by the notochord. The amniotic cavity develops around the neural tube, with the aortae appearing on either side.

As the neural tube closes, the aortae draw closer, as they do not grow at the same rate. This approximation reorganises the longitudinal axis of the embryo, creating a space between the **somatopleure** and the **splanchnopleure**. At the same time, the digestive tract develops, and fluid begins to integrate into a defined space.

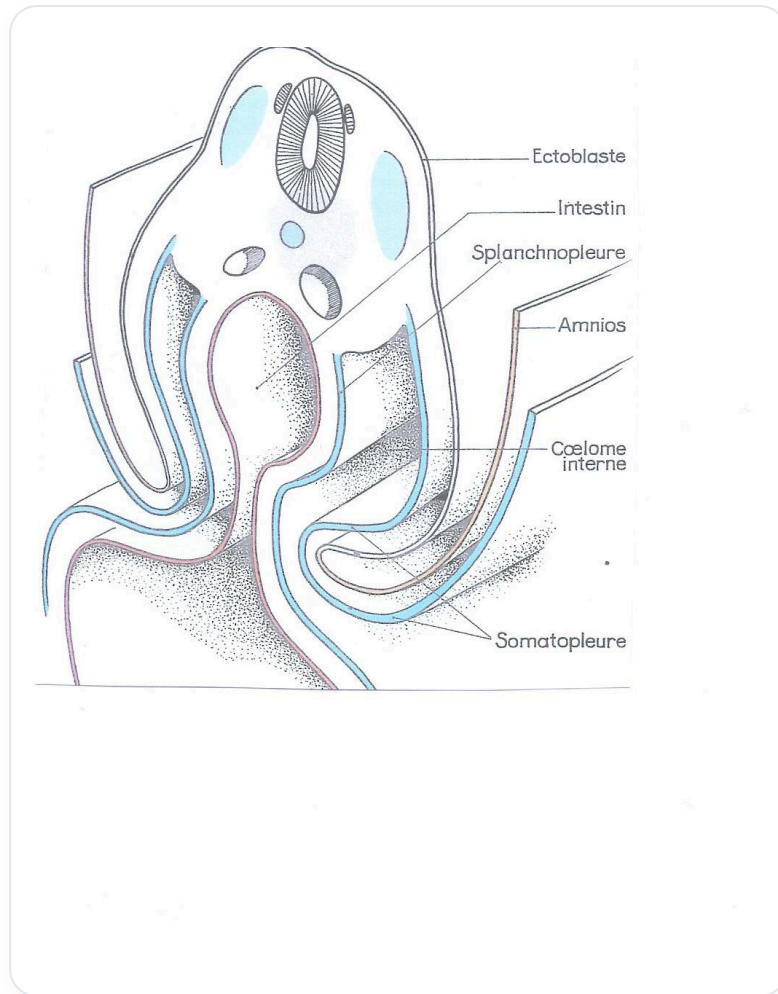


FIG. — *Embryonic coelom development*

It is crucial to visualise this growth process. The amniotic cavity progressively envelops the space, like a balloon. This movement, orchestrated by the aortae and the notochord, allows the integration of the external coelom into an internal coelom.

At an advanced stage, we observe the integration of the external coelom into an internal coelom, illustrating that the digestive tract is not only filled with bacteria but also constitutes a **genome**. The latter contains information about the environment and even genetic elements, forming a **holobiont** that integrates this genetic environment. This process will be energised at birth, preparing the flora that will play a crucial role in development.

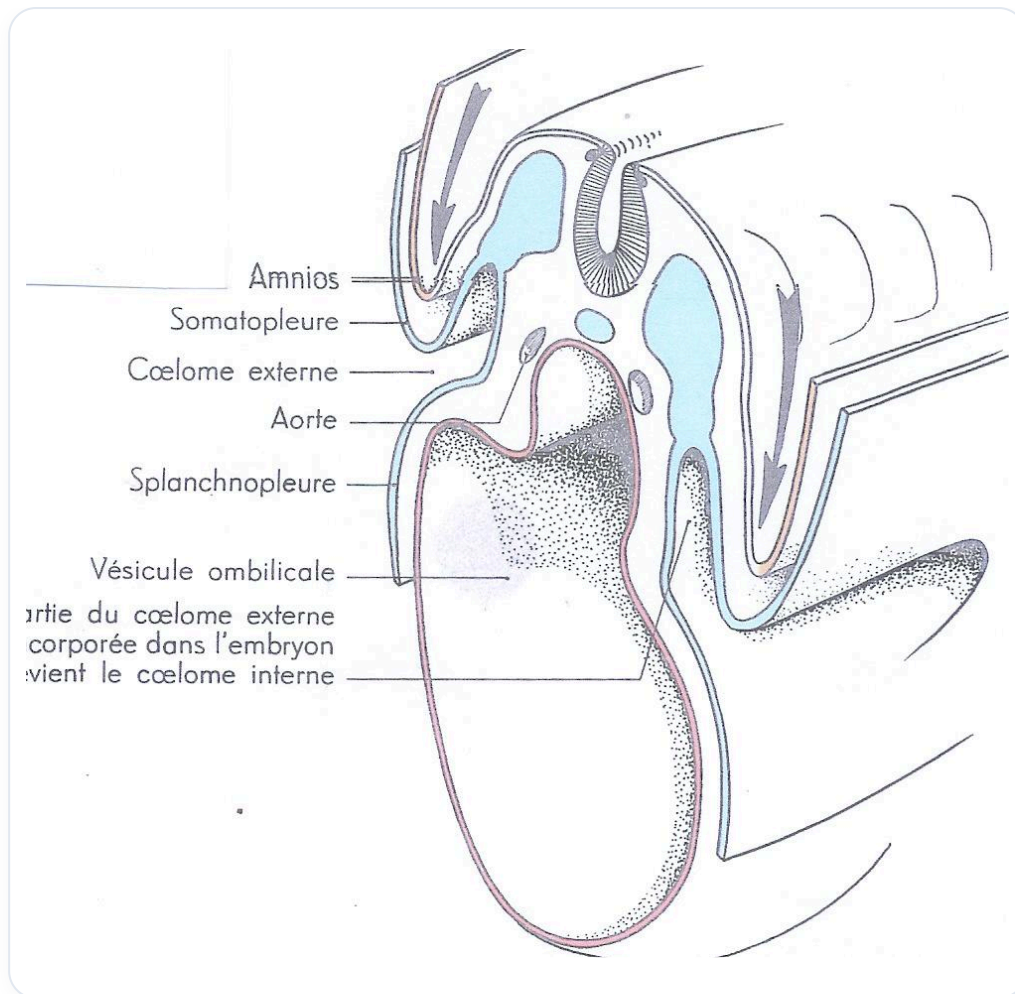


FIG. — *Lateral embryonic folding*

The genes present in the flora interact with the brain and influence thought, highlighting the importance of an adequate environment for evolution. The environment in which we evolve also impacts our genetics and epigenetics, which are essential for our adaptability.

The first image of the peritoneal cavity shows a peritoneum that has **metabolic**, **protective**, and **mechanical** functions. It participates in motility and mobility, being integrated by motility. Its origin is the **mesoderm**, which gives it support and mobility functions. Specialised **mesothelial** cells at the level of the mesoderm allow the peritoneum to capture inter- and intra-peritoneal information. This serous membrane is very large, measuring approximately 2 square metres.

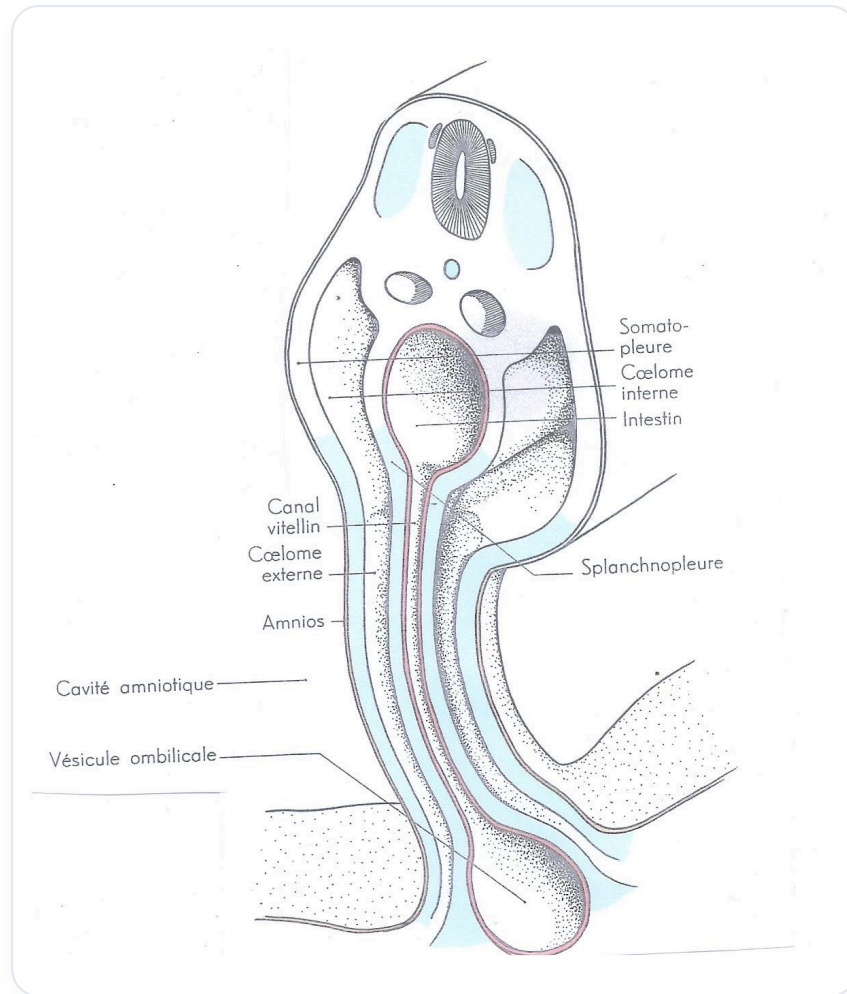


FIG. — *Embryo, coelom, amnion*